

## Assignment - 2: NumPy

1. Create a 3x3 matrix array with values ranging from 2 to 10.

### PROGRAM

```
import numpy as np

a=np.arange(2,11,1)
b=np.reshape(a, (3, 3))

print(b)
```

### OUTPUT

```
[[2  3  4]
 [5  6  7]
 [8  9 10]]
```

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2. Create a NumPy array having user input values and convert the integer type to the float type of the elements of the array.

For instance:

Original array [1, 2, 3, 4]

Array converted to a float type: [ 1. 2. 3. 4.]

### PROGRAM

```
import numpy as np

y=[]
a=int(input("Size of array:"))

for i in range(a):
    x=(int(input("Element:")))
    y.append(x)

c=np.array(y)
print("Original array is:",c)

z = np.asarray(c, dtype = float)
print("Converted array is:",z)
```

### INPUT

```
Size of array:4
Element:1
Element:2
Element:3
Element:4
```

### OUTPUT

```
Original array is: [1 2 3 4]
Converted array is: [1. 2. 3. 4.]
```

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3. Write a NumPy program to append values to the end of an array.

For instance:

Original array: [10, 20, 30]

After append values to the end of the array: [10 20 30 40 50 60 70 80 90]

**PROGRAM**

```
import numpy as np
```

```
x = [10, 20, 30]
```

```
print("Original array:",x)
```

```
x = np.append(x, [[40, 50, 60], [70, 80, 90]])
```

```
print("After append values to the end of the array:",x)
```

**OUTPUT**

Original array: [10, 20, 30]

After append values to the end of the array: [10 20 30 40 50 60 70 80 90]

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4. Create two NumPy arrays and add the elements of both the arrays and store the result in sumArray.

**PROGRAM**

```
import numpy as np
```

```
x=np.array([10,20,30])
```

```
y=np.array([40,50,60])
```

```
sumArray = np.add(x,y)
```

```
print("Array of addition of elements:",sumArray)
```

**OUTPUT**

Array of addition of elements: [50 70 90]

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5. Create a 3\*3 array having values from 10-90(interval of 10) and store that in array1. Perform the following tasks:

- Extract the 1<sup>st</sup> row from the array.
- Extract the last element from the array.

**PROGRAM**

```
import numpy as np
```

```
a=np.arange(10,100,10)
```

```
array1=np.reshape(a, (3, 3))
```

```
print("First row of the array is:",array1[0,:])
```

```
print("Last element of the array is:",array1[-1,-1])
```

**OUTPUT**

First row of the array is: [10 20 30]

Last element of the array is: 90

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